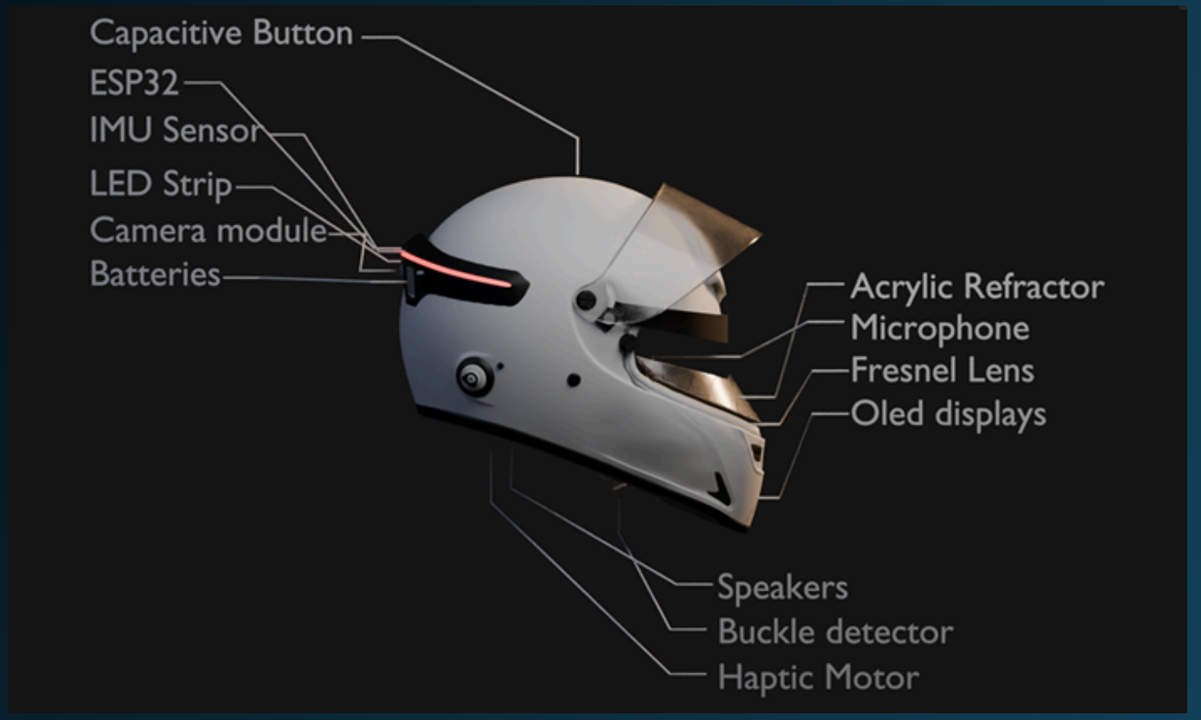
HELMET DESIGN & HARDWARE

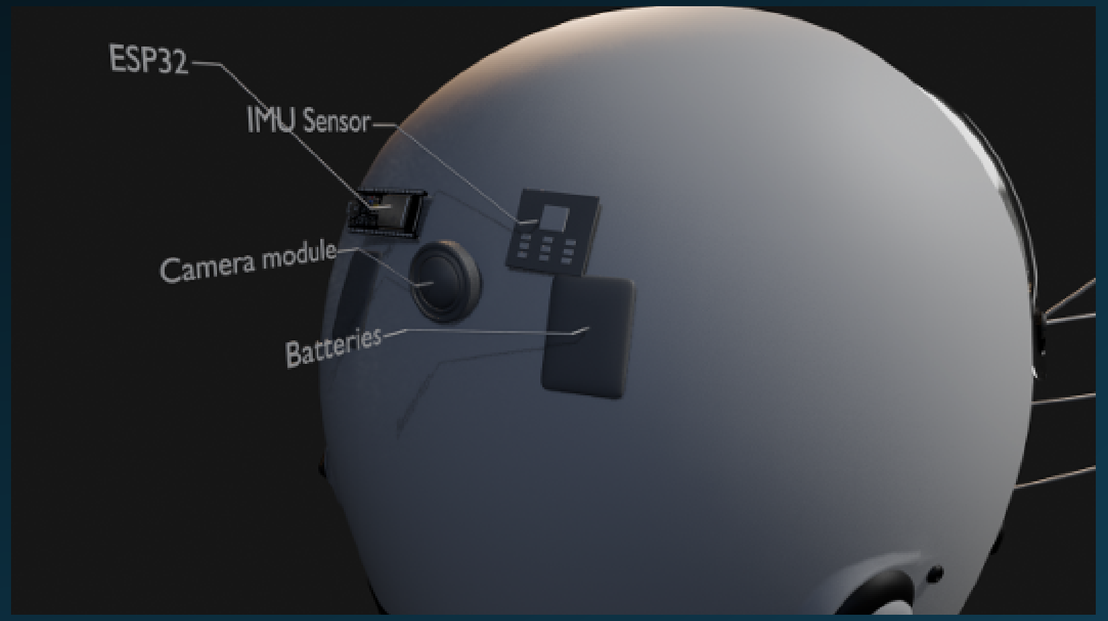
Inputs

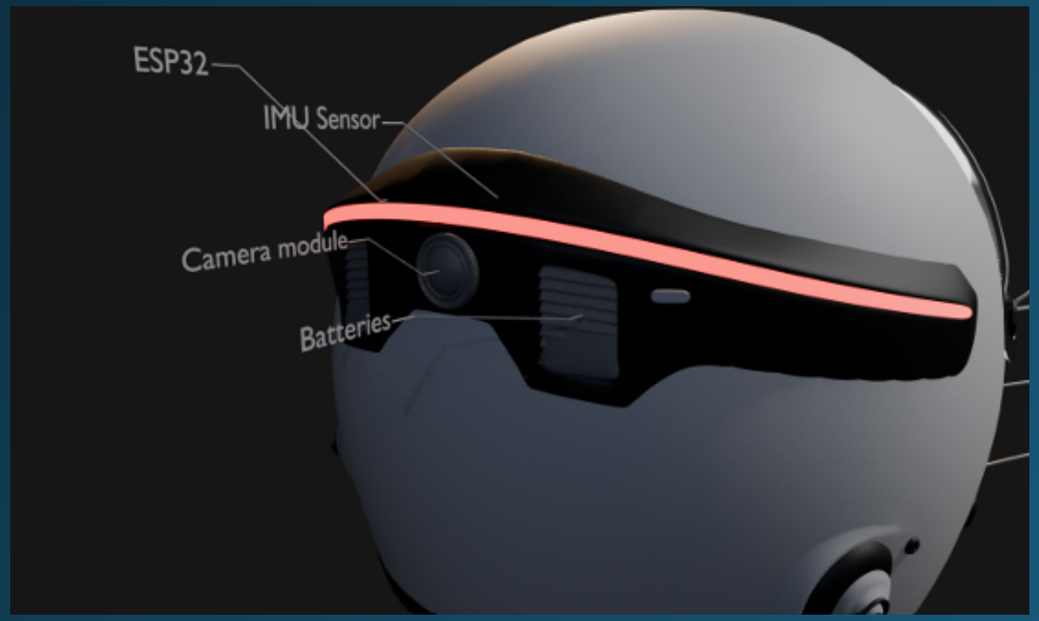
- 9-axis IMU
- Camera
- Microphone
- Hall effect buckle sensor
- Capacitive wear detection

Outputs

- HUD displays
- Stereo speakers
- Haptic motors
- LED strips


Capacitive Button
ESP32
IMU Sensor
LED Strip
Camera module
Batteries
Acrylic Refractor
Microphone
Fresnel Lens
Oled displays
Speakers
Buckle detector
Haptic Motor


ESP32
IMU Sensor
Camera module
Batteries


ESP32
IMU Sensor
Camera module
Batteries

DATA GENERATION

Data Types

- Motion data (IMU)
- Distance data (rear & side sensors)
- Environmental data (rain, humidity, light, temperature)
- Audio and voice input
- GPS and navigation data (from phone)

Data Frequency

- IMU & distance sensors: real-time
- Environmental sensors: periodic
- GPS & weather: event-based

FLOW ARCHITECTURE

